Formalism: recovery limits, EFE, and tempered aggregation

Formalism: recovery limits, EFE, and tempered aggregation I

The primitives of sec. 5 are governed by a compact set of machine-checkable identities. Their counter is monotone across the methods and this section: Definitions 1 and 2 (posterior and cavity / PVI update, sec. 5.3); Lemma 3 (Rényi KL limit, sec. 5.6); Proposition 4 (β-loss and rcce NLL limits, sec. 5.7); Theorem 5 and Corollary 6 (belief-sharing and Bayes recovery, sec. 5.8); and Proposition 1 below (expected-free-energy identity). Each carries a tested residual. None grants a bounded-influence guarantee to the server-side `robust_aggregate` heuristic; that boundary is stated in sec. 3.3.

Recovery limits as the proof surface

Recovery limits as the proof surface I

The recovery limits separate client and server claims. The divergence and loss limits recover the standard-Bayes client update; the independently tested `robust_aggregate(robustness=0) == log_linear_pool` identity recovers the project's standard server pool. Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions in sec. 5.8, that pool is a categorical specialization of Eq. 7's message-combination term, not recovery of the complete source protocol (Friston et al. 2024). We collect the five residuals that pin those limited claims, each emitted by the test-suite and reported in sec. 7.1, never hardcoded. Read each row of the table below as a triple: the robust primitive, the trusting knob value at which it must collapse onto its standard-Bayes or project-local counterpart, and the tested residual measuring whatever gap survives at that value. The rows differ in what kind of check they

Recovery limits as the proof surface I

are. The divergence and loss rows evaluate inside the implementation's closed-form switch band, so their zeros are exact branch identities — guaranteed by construction, not measurements that could have come out otherwise. Their genuine falsifiers are the *off-switch* convergence residuals, evaluated just outside the band (at $\alpha = 1.00001$, $\beta = 1e-06$, $q_{\mathrm{loss}} = 1.00 \times 10^{-6}$) where the general formulas run and a nonzero gap is possible: those residuals are 1.66 $\times$10^{-5}, 1.24 $\times$10^{-5}, and 1.12 $\times$10^{-5} respectively, and any failure of those quantities to shrink toward the limit would falsify the containment claim. The posterior row is a measured identity on the general code path, so its near-zero residual is itself the falsification surface. The aggregate row's zero at exactly $c = 0$ is likewise branch-exact; the identity is additionally exercised on the iterative code path at near-zero robustness, where the consensus must still land on the log-linear pool to tight tolerance.

Recovery limits as the proof surface I

Table 1: Recovery residuals: the largest observed discrepancy between each robust primitive and its standard-Bayes limit, over the recovery band. Each is a maximum absolute difference in the natural units of the quantity (pmf entries for the posterior and aggregate rows, nats for the divergence and loss rows); the aggregate, divergence, and loss rows are exactly zero (bit-identical) and the posterior row is exact to machine precision (about one ULP), so the limits are verified identities, not approximations. The labeled presentation of these residuals lives in sec. 7.1.

Identity (owner statement)	Trusting limit	Tested residual
Rényi $\to$ KL (Lemma 3, eq. 3)	$\alpha \to 1$	0
rcce $\to$ NLL (Proposition 4, eq. 5)	$q_{\text{loss}} \to 0$	0
β-loss $\to$ NLL (Proposition 4, eq. 4)	$\beta \to 0$	0
generalized posterior $\to$ Bayes (Corollary 6, eq. 6)	KL, NLL	5.55e-17
`robust_aggregate` $\to$ log-linear pool (Theorem 5, eq. 7)	$c = 0$	0

The central project identity is the aggregation collapse of eq. 7: `robust_aggregate(robustness=0)` equals the log-linear pool eq. 6. Its source bridge is deliberately narrow. Take a finite common support with $q_n(s) > 0$ and represent each Eq. 7 softmax input as a posterior log potential $m_n(s) = \log q_n(s) + \kappa_n$, with κ_n constant in s and fixed declared weights w_n independent of the emerging consensus. Softmax then cancels the additive constants and yields the project pool. This identifies only the source equation's message-combination term; it does not identify source message construction, cavity/exclusion policy, scheduling, generative factors, or the complete protocol. Theorem 5 (sec. 5.8) states that specialization and the local $c = 0$ identity; the residual 0 above pins the latter. Corollary 6 establishes the separate client result: `generalized_posterior(KLD, NLL)` reproduces the closed-form prior-times-likelihood Bayes posterior of eq. 6 to residual 5.55e-17. Pooling such local posteriors has the stated categorical specialization only under the theorem's assumptions. The honesty contract binds here: the theorem and corollary cover only the recovery identity and the per-agent rigorous axis (Proposition 4); no statement transfers the bounded-influence guarantee to the server-side divergence-reweighting heuristic, whose positive property is the $\texttt{robustness} = 0$ limit of eq. 7. A scoped no-go rejects a declared separable objective class without certifying another.

Expected-free-energy identity as an algebraic check

Expected-free-energy identity as an algebraic check I

The active-inference substrate that drives the studies of sec. 7 is a categorical specialization of the expected-free-energy algebra discussed by Friston et al. (Friston et al. 2024). It decomposes the expected free energy of a policy π into two equivalent two-term forms. The risk-plus-ambiguity (cost) view and the negated pragmatic-plus-epistemic (value) view are the same scalar $G(\pi)$ rearranged (Da Costa et al. 2020; Friston et al. 2024):

Expected-free-energy identity as an algebraic check I

$$G(\pi) \;=\; \underbrace{\text{risk} + \text{ambiguity}}_{\text{cost view}} \;=\; -\big(\underbrace{\text{pragmatic} + \text{epistemic}}_{\text{value view}}\big). \tag{1}$$

The two views are not approximations of one another; they are the same scalar rearranged. In the implementation the shared entropy term enters both sides of the rearrangement, so the identity residual is zero by construction — it is a definitional consistency check on the decomposition's bookkeeping, not an independent measurement. The scientific content lives in the per-term semantics, which are pinned independently of the identity (see the closing clause of the proposition below):

Expected-free-energy identity as an algebraic check I

$$(\text{risk} + \text{ambiguity}) \; + \; (\text{pragmatic} + \text{epistemic}) \; \equiv \; 0. \tag{2}$$

Proposition (Expected-free-energy decomposition identity)

For the categorical generative model of `expected_free_energy.py`, the cost decomposition and the negated value decomposition of (1) yield the same $G(\pi)$, so the identity (2) holds with a residual at machine precision. The risk term is $\mathrm{KL}(q(o \mid \pi) \,\|\, p_C(o))$, the ambiguity term is the expected likelihood entropy $\mathbb{E}_{q(s)}[H[p(o \mid s)]]$, the pragmatic value is the expected log-preference $\mathbb{E}_{q_\pi(o)}[\ln p_C(o)]$, and the epistemic value is the state-outcome mutual information $H[q_\pi(o)] - \mathbb{E}_{q(s)}[H[p(o \mid s)]]$; the identity follows from the cross-entropy split of the risk and the entropy split of the epistemic value. The residual of the decomposition is pinned to zero at a tolerance of 10^{-9}, and each term's semantics is pinned independently (deterministic likelihoods give zero ambiguity; uninformative likelihoods give zero epistemic value; preference-matched predictions lower risk).

The executed formal-specialization diagnostic of fig. 1 uses a uniform prior over the nine locations. This is intentional: the canonical sentinel-world D_0 is a point mass at the den, and under that fully resolved prior the mutual-information term is zero because there is no state uncertainty for an observation to reduce. The uncertainty-bearing diagnostic makes the epistemic term visible without changing the canonical D_0 used by the inference and recovery studies. Thus a near-zero epistemic value is a meaningful null condition, not a missing term. fig. 1 shows the additive risk-plus-ambiguity view beside a signed pragmatic/epistemic waterfall whose terminal endpoint is $G(\pi)$, labels epistemic value as $I(s; o \mid \pi)$, and annotates the identity residual; it visualizes Proposition 1, not a fitted

Expected-free-energy identity as an algebraic check I

result, so it carries no error bars.

Expected-free-energy identity as an algebraic check I

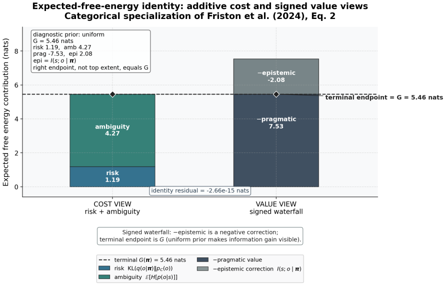

Figure 1: Expected-free-energy decomposition for the categorical generative model (expected_free_energy.py). Source relation: formal specialization of Friston et al. (2024), Eq. 2; estimand: categorical EFE identity in nats. x-axis: two views of the same identity (left, additive cost view: risk + ambiguity; right, signed value waterfall: positive minus-pragmatic contribution followed by a negative epistemic correction). y-axis: EFE contribution in nats. The heavy endpoint marker and connector, rather than the intermediate top extent, identify the terminal $G(\pi)$ value. The epistemic term is state–outcome mutual information $I(s; o \mid \pi)$; it is visible because the diagnostic prior is uniform, whereas the canonical point-mass D_0 is the corresponding zero-information null. The finite terms satisfy the identity at machine precision. This deterministic algebraic check has no error bars or independent sample size, and it does not reproduce every parameter-learning term in the source equation.

Expected-free-energy identity as an algebraic check I

The expected-free-energy identity of eq. 2 is the action-selection counterpart of the inference-side recovery limits collected above: both are exact, closed-form, machine-checkable identities over the same categorical generative model. Together they establish that the FedGVI-federated active-inference colony of this work is built on verified algebra throughout — the per-agent generalized-Bayes update recovers standard Bayes, the aggregation recovers the project log-linear pool under its qualified categorical bridge, and the policy scoring decomposes exactly — so every robustness result in sec. 7 is a controlled departure from a known, tested fixed point rather than an unmoored claim.

Tempered aggregation free energy and the accuracy-guarantee trade

The recovery limits and the expected-free-energy identity fix the *endpoints* of the aggregation family; the remaining formal question is what a controlled departure from the unit-entropy server buys and what it costs. The objective-backed variational aggregator of sec. 11 holds its consensus-entropy term at unit weight. Freeing that single coefficient produces a one-parameter family whose only moving part is the sharpness of the consensus, and whose raw effective-weight bound is provably untouched. The following proposition isolates exactly that separation — algebra that moves versus algebra that does not — before the interpretation subsections turn to the empirical accuracy question the algebra cannot settle on its own.

Proposition (Tempered aggregation free energy)

Let $\lambda > 0$ be an entropy weight and F_λ the objective of equation 25. For a fixed effective-weight vector a, the q-block minimizer of F_λ is the tempered-softmax update in equation 26. At $\lambda = 1.0$ the objective, both block updates, and the endpoint-selection rule reduce to the standard variational aggregate (Definition 10, Section 11) bit-for-bit. The a-block update and its raw effective-weight bound $a_n \leq w_n$ contain no λ and are therefore unchanged for every $\lambda > 0$.

The proposition is intentionally more specific than the phrase "temperature improves robustness." It identifies exactly which part of the variational server changes when the entropy coefficient changes, and it separates that algebra from the empirical accuracy question. The objective and its update rules are a generalized-Bayes construction in the sense of (Bissiri et al. 2016; Knoblauch et al. 2022), while the particular client/server decomposition is the one implemented and tested here.

What the entropy weight controls I

For a fixed effective-weight vector a and $\lambda > 0$, the q-block in eq. 26 is a weighted geometric pool of the local posteriors with inverse temperature $1/\lambda$. Lower λ concentrates more sharply on states that receive consistent log-belief support; larger λ spreads probability mass and retains more entropy. As $\lambda \downarrow 0$, the positive-temperature expression approaches a winner-take-most consensus (subject to ties and finite numerical support), whereas large λ approaches a flatter distribution. The implementation exposes that endpoint as a separately defined deterministic tied-argmax rule; it does not substitute $\lambda = 0$ into the objective or coordinate update. This is a controlled change in the consensus geometry, not an automatic outlier detector.

The coupling matters. Although the formula for the a-block does not contain λ, the fixed point can still change because a_n is evaluated at the new q. The correct statement is therefore conditional: for any current consensus, the effective-weight update and the raw bound $a_n \leq w_n$ are unchanged; after alternating updates, different temperatures can reach different coupled (q, a) fixed points. This distinction prevents the temperature result from being read as a theorem that the normalized influence or accuracy is invariant in λ.

Recovery at the qualified log-linear-pool corner I

At the configured default $\lambda = 1.0$, the entropy coefficient is the unit coefficient used by the original variational aggregator. The implementation therefore recovers that aggregator bit-for-bit, including its block updates and its endpoint-selection rule. Turning the robustness strength c to zero then sets every server weight to its base value and gives the tempered log-linear pool. At the default temperature this is the ordinary log-linear pool of eq. 6. Under the shared-support, posterior-log-potential, and fixed-weight assumptions of sec. 5.8, that is the categorical specialization of Eq. 7's message-combination term; it is not the complete belief-sharing protocol of (Friston et al. 2024). Away from that default, the result is a tempered generalization of the pool and should not be described as Friston's Eq. 7 itself.

This nested limit is useful for interpretation. The $c \to 0$ limit identifies the aggregation family with a known consensus operator; the $\lambda = 1.0$ slice identifies the objective-backed implementation used in the main server comparison. Neither limit grants the server-side `robust_aggregate` heuristic a variational objective or a bounded-influence guarantee. The FedGVI literature's rate and robustness results remain attached to their stated loss, divergence, and sampling assumptions (Mildner, Hamelijnck, et al. 2025; Mildner, Giampouras, et al. 2025).

What the accuracy–guarantee trade can establish I

The executed grid over $\lambda \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1\}$ is a finite sensitivity study, not a search over a continuous optimum. It asks whether any tested temperature narrows the point-accuracy gap to the sharp heuristic while retaining the same effective-weight update. The closest tested temperature is $\lambda^* = 0.3$, with observed gap 0.0008. These tokens are computed from the executed contaminated-colony trials and are reported with the grid definition so a reader can reproduce the selection rule.

What the accuracy–guarantee trade can establish I

The result has two distinct readings. If a lower temperature improves the paired point-accuracy comparison, it provides design evidence that entropy regularization can be tuned rather than accepted as a fixed conservatism penalty. If no tested temperature closes the gap, that negative result is still informative: within this objective family, the same entropy mechanism that keeps consensus diffuse can limit exact point recovery under confident contamination. In either case, the grid does not identify a universally best temperature, establish minimax robustness, or transfer the variational raw weight bound to a different estimator. Generalized-Bayes calibration and robust-loss theory motivate the family, but only the executed categorical design supports the present finite-grid statement (Bissiri et al. 2016; Knoblauch et al. 2022; Mildner, Hamelijnck, et al. 2025).

Publication-facing interpretation I

The practical contract is consequently three-part. Use the default temperature when exact compatibility with the tested variational server is the priority. Explore the declared grid when the application can trade concentration against the same raw effective-weight control. Treat any selected temperature as a configuration-specific empirical choice until it is tested under a new contamination mechanism, colony size, sensor model, or loss. This is the appropriate bridge between the formal objective and the active-inference setting: it exposes a tunable consensus geometry while keeping recovery, guarantee, and accuracy claims on separate evidence tracks.